DEVELOPMENT OF WEB APPLICATIONS

Individual In-Depth Study – Web Framework Exploration (1.5 ECTS)

Assignment	Details
Form	Individual
Credits	1.5 credits (hp)
Examination parts	Report: Individual in-depth study with implementation (1.0 ECTS) Seminar: individual presentation (0.5 ECTS)
Grade	Fail/Pass (U/G)
Assessed Learning outcomes	2.1, 2.2, 3.1
Submission	Canvas; dates and seminar arrangements are announced there

1. Purpose & Alignment with Laboratory Assignment 2

The purpose of the individual in-depth study is to provide you with an advanced theoretical and practical understanding of a selected modern web framework, as well as rigorous training in academic writing according to the IMRAD structure (Introduction, Methods, Results, Discussion).

To establish a cohesive narrative across the course, the study directly builds upon your experience from Laboratory Assignment 2. In Lab 2, you work in pairs to construct a database-driven web application using C#, ASP.NET Core MVC, and Entity Framework Core. In this in-depth study, you work individually to explore a different web framework. You are strongly encouraged to reuse the same application domain and data model (e.g., booking system, item lending, or PlantSwap) defined in Lab 2. This eliminates the overhead of defining new requirements and allows you to focus on comparing how your chosen framework solves the exact same problem compared to ASP.NET Core MVC.

Why is a practical code implementation required?

1. Syllabus Requirement: The formal course syllabus explicitly designates this 1.0 ECTS module as 'Individual in-depth study with implementation'.
2. Individual Assessment: Lab 2 is executed in pairs. The in-depth study represents your individual opportunity to demonstrate competence in practical skill targets (2.1 & 2.2).
3. Empirical Basis for IMRAD: Hands-on development provides the empirical data required for your Results and Discussion sections regarding Developer Experience (DX), state management, component architecture, and routing.

2. Framework Selection & Practical Micro-Artifact

You must individually select an established web framework for specialization.

- **Allowed Frameworks:** Examples include React, Angular, Vue.js, Svelte, Django, Ruby on Rails, Express.js, or Flask. (Note: ASP.NET Core is excluded as it is covered in Lab 2).
- **Approval:** Your choice of framework must be registered and approved by Malin Nilsson.
- **Scope of Practical Implementation (Micro-Artifact):** To ensure a reasonable workload (approx. 4–8 hours of coding), you are NOT expected to build a complete application with full database integration, identity/security, and deployment. Instead, you will create a focused micro-artifact that demonstrates core framework mechanisms. You may choose one of the following two tracks:

Track	Implementation Type	Scope & Lab 2 Integration
Track A	Frontend SPA (React, Vue, Angular, Svelte)	Build 2–3 interactive UI components (e.g., list view, filter, detail/form view) that consume Web API endpoints created in your Lab 2 backend.
Track B	Standalone Micro-Prototype (Django, Rails, Express, Flask)	Build a minimal standalone web prototype implementing CRUD operations for a single core entity from your Lab 2 domain.

3. Theoretical Analysis & Comparative Research Question

In parallel with your micro-artifact, you will conduct a theoretical and empirical analysis evaluating the chosen framework. Your analysis must synthesize academic/industry literature with your practical observations from both Lab 2 and the in-depth study project.

Mandatory Comparative Research Question:

"What are the architectural advantages and disadvantages of [Selected Framework], and how do its component model, state management, and developer experience differ from ASP.NET Core MVC as applied in Laboratory Assignment 2?"

4. Report Structure (IMRAD)

The written report must strictly follow the IMRAD format. A standardized template will be available on Canvas. The report must contain the following main sections:

Title Page & Abstract

Concise title and a structured abstract of max 250 words summarizing purpose, methods, key results, and conclusions.

1. Introduction

- Background & Problem Statement: Introduction to the selected framework and its role in modern web architecture.
- Purpose & Research Question: Explicit statement of study objectives and the research question.
- Target Audience & Significance: Relevance for developers and software engineers.
- Literature Review: Theoretical foundation based on official documentation and peer-reviewed/industry literature.

2. Methods

- Study Object & Scope: Technical specification of the chosen framework and micro-artifact domain.
- Literature Search Procedure: Strategy for source selection and inclusion criteria.
- Implementation & Evaluation Protocol: Description of micro-artifact development and comparative evaluation criteria against ASP.NET Core MVC.

3. Results

- Micro-Artifact Overview: Brief description of the practical implementation with code snippets and architecture diagrams.
- Comparative Technical Analysis: Structured evaluation of framework features, performance, DX, and state handling compared to ASP.NET Core MVC (include comparative tables).

4. Discussion & Conclusion

- Interpretation & Reflection: Critical discussion of findings in relation to literature and Lab 2 experience.
- Limitations & Developer Experience: Reflection on study limitations, learning curve, and trade-offs.
- Conclusion: Concise, direct answer to the research question.

References

Comprehensive reference list formatted according to the Harvard citation style.

5. Examination & Submission Requirements

1. Written Report & Micro-Artifact (1.0 ECTS): Submit the final PDF report written in IMRAD format alongside a repository link (e.g., GitHub) containing the micro-artifact source code via Canvas.

2. Examination Seminar (0.5 ECTS): Individual oral presentation (approx. 10 minutes) including a live demonstration of the micro-artifact, followed by peer discussion and Q&A.